

STATE QUOTIENTS AND AGENT-PERMUTATION SHARING FOR MARKOV POTENTIAL GAMES

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1 PROBLEM SETUP

For a finite set $\mathcal{X}$, let $|\mathcal{X}|$ denote its cardinality and $\Delta(\mathcal{X})$ the probability simplex over it. We write $[n] := \{1, \dots, n\}$.

1.1 FINITE DISCOUNTED MARKOV GAMES

A finite discounted Markov game is

$$M = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, P, \vec{r}, \mu, \gamma \rangle.$$

The agent set is $\mathcal{N} = [N]$, the state space is $\mathcal{S}$, and the joint action space factors as

$$\mathcal{A} = \mathcal{A}^1 \times \dots \times \mathcal{A}^N.$$

For agent i , write $\mathcal{A}^{-i} := \prod_{j \neq i} \mathcal{A}^j$. We write a joint action as $a = (a^1, \dots, a^N)$ and as $a = (a^i, a^{-i})$ when focusing on agent i . The transition kernel is $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, the initial distribution is $\mu \in \Delta(\mathcal{S})$, and the discount factor is $\gamma \in [0, 1]$. Each normalized reward function is $r^i : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$.

All policies are stationary product policies $\pi = (\pi^1, \dots, \pi^N)$, where

$$\pi^i : \mathcal{S} \rightarrow \Delta(\mathcal{A}^i), \quad \pi(a | s) = \prod_{i=1}^N \pi^i(a^i | s).$$

For agent i , define

$$\begin{aligned} G_t^i &:= \sum_{\ell=0}^{\infty} \gamma^\ell r^i(s_{t+\ell}, a_{t+\ell}), \\ V_\pi^i(s) &:= \mathbb{E}_\pi[G_t^i | s_t = s], \\ Q_\pi^i(s, a) &:= \mathbb{E}_\pi[G_t^i | s_t = s, a_t = a], \\ A_\pi^i(s, a) &:= Q_\pi^i(s, a) - V_\pi^i(s), \\ V_\pi^i(\mu) &:= \mathbb{E}_{s \sim \mu}[V_\pi^i(s)]. \end{aligned}$$

For a local action $b \in \mathcal{A}^i$, the marginalized local advantage is

$$A_\pi^i(s, b) := \sum_{a^{-i} \in \mathcal{A}^{-i}} \pi^{-i}(a^{-i} | s) A_\pi^i(s, b, a^{-i}). \quad (1)$$

The unnormalized discounted state-visitation measure is

$$d_\mu^\pi(s) := \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s | s_0) \right], \quad \sum_{s \in \mathcal{S}} d_\mu^\pi(s) = \frac{1}{1 - \gamma}.$$

1.2 MARKOV POTENTIAL GAMES

Definition 1 (Markov potential game). A Markov game is a *Markov potential game* (MPG) if there is a function $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ such that, for every agent i , state s , opponents' policy π^{-i} , and pair of

local policies $\pi^i, \tilde{\pi}^i$,

$$\begin{aligned} & V_{\tilde{\pi}^i, \pi^{-i}}^i(s) - V_{\pi^i, \pi^{-i}}^i(s) \\ &= \mathbb{E}_{(\tilde{\pi}^i, \pi^{-i})} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right] - \mathbb{E}_{(\pi^i, \pi^{-i})} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right]. \end{aligned}$$

For a product policy π , define its total potential by

$$\begin{aligned} \Phi_{\pi}(s) &:= \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| s_0 = s \right], \\ \Phi_{\pi}(\mu) &:= \mathbb{E}_{s \sim \mu} [\Phi_{\pi}(s)]. \end{aligned}$$

The MPG condition is equivalently

$$V_{\tilde{\pi}^i, \pi^{-i}}^i(s) - V_{\pi^i, \pi^{-i}}^i(s) = \Phi_{\tilde{\pi}^i, \pi^{-i}}(s) - \Phi_{\pi^i, \pi^{-i}}(s).$$

A fully cooperative game with $r^i \equiv r$ for all agents is an MPG with $\phi = r$.

1.3 NASH GAP AND REGULARITY

Definition 2 (Nash gap and ϵ -Nash policy). The *Nash gap* of a product policy π is

$$\text{Nash gap}_M(\pi) := \max_{i \in \mathcal{N}} \left(\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right),$$

where the inner maximum ranges over all stationary local policies $\tilde{\pi}^i : \mathcal{S} \rightarrow \Delta(\mathcal{A}^i)$. A policy is ϵ *Nash* if $\text{Nash gap}_M(\pi) \leq \epsilon$.

Assumption 3 (Standing regularity).

(i) *Positive visitation*. For every product policy π and state s , $d_{\mu}^{\pi}(s) > 0$. Consequently,

$$\kappa := \max_{\pi, \pi'} \left\| d_{\mu}^{\pi} / d_{\mu}^{\pi'} \right\|_{\infty} = \max_{\pi, \pi'} \max_{s \in \mathcal{S}} \frac{d_{\mu}^{\pi}(s)}{d_{\mu}^{\pi'}(s)} < \infty. \quad (2)$$

(ii) *Bounded total potential*. There are finite constants $\Phi_{\min} \leq \Phi_{\max}$ such that

$$\Phi_{\min} \leq \Phi_{\pi}(s) \leq \Phi_{\max} \quad \text{for every product policy } \pi \text{ and state } s.$$

The stationary product-policy space is a finite product of probability simplexes and is therefore compact. The map $\pi \mapsto d_{\mu}^{\pi}(s)$ is continuous for every state s . Consequently, positive visitation for every policy implies a uniformly positive minimum visitation over policies and states, which justifies the finiteness of κ in equation 2.

2 FINITE-GROUP STATE SYMMETRY

Let G be a finite group with identity e . A left action of G on a finite set $\mathcal{X}$ is a family of bijections $\{T_g : \mathcal{X} \rightarrow \mathcal{X}\}_{g \in G}$ satisfying

$$T_e = \text{id}, \quad T_g \circ T_h = T_{gh}.$$

The orbit and stabilizer of $x \in \mathcal{X}$ are

$$\begin{aligned} [x] &:= \{T_g[x] : g \in G\}, \\ \text{Stab}(x) &:= \{g \in G : T_g[x] = x\}. \end{aligned}$$

The orbits partition $\mathcal{X}$ and form the orbit space $\mathcal{X}/G$. The orbit-stabilizer theorem gives

$$|[x]| = \frac{|G|}{|\text{Stab}(x)|} \leq |G|, \quad |\mathcal{X}/G| \geq \frac{|\mathcal{X}|}{|G|}.$$

Definition 4 (Finite-group state symmetry). Let G act on $\mathcal{S}$ through bijections $\{L_g\}_{g \in G}$. A Markov game M is G -state-symmetric if, for every $g \in G$, $s, s' \in \mathcal{S}$, $a \in \mathcal{A}$, and $i \in \mathcal{N}$,

$$\begin{aligned} P(L_g[s'] \mid L_g[s], a) &= P(s' \mid s, a), \\ r^i(L_g[s], a) &= r^i(s, a). \end{aligned} \quad (3)$$

The action a is unchanged: the group acts only on the state coordinate.

Assumption 5 (Invariant potential and initial distribution). For the G -state-symmetric MPG M , assume

$$\begin{aligned} \phi(L_g[s], a) &= \phi(s, a), \\ \mu(L_g[s]) &= \mu(s) \end{aligned}$$

for every $g \in G$, $s \in \mathcal{S}$, and $a \in \mathcal{A}$.

Potential invariance is automatic in a cooperative game when $\phi = r$. Initial-distribution invariance means that states in the same orbit receive equal initial mass.

Definition 6 (G -invariant policy). A product policy π is G invariant if

$$\pi^i(a^i \mid L_g[s]) = \pi^i(a^i \mid s) \quad \text{for every } i, g, s, \text{ and } a^i.$$

Equivalently, $\pi(a \mid L_g[s]) = \pi(a \mid s)$ for every joint action a .

The per-agent conditions imply joint invariance by multiplying the local action probabilities. Conversely, joint invariance implies each per-agent condition after summing over the other agents' actions. Thus an invariant policy assigns the same local action distribution to every state in an orbit. We use the same terminology for a local policy: ν^i is G invariant if $\nu^i(a^i \mid L_g[s]) = \nu^i(a^i \mid s)$ for every g, s, a^i .

3 THE QUOTIENT MARKOV GAME

Let

$$\bar{\mathcal{S}} := \mathcal{S}/G$$

be the set of state orbits, and let

$$\rho : \mathcal{S} \rightarrow \bar{\mathcal{S}}, \quad \rho(s) := [s],$$

be the orbit map. Fix a representative $\iota(\bar{s}) \in \bar{s}$ for each $\bar{s} \in \bar{\mathcal{S}}$.

Definition 7 (Quotient Markov game induced by G). Let M be a G -state-symmetric Markov game. Its *quotient Markov game* induced by G is

$$\bar{M} := \langle \mathcal{N}, \bar{\mathcal{S}}, \mathcal{A}, \bar{P}, \bar{r}, \bar{\mu}, \gamma \rangle.$$

It has the same agents, local action spaces, and discount factor as M . For any $s \in \bar{s}$, its transition kernel, rewards, and initial distribution are

$$\bar{P}(\bar{s}' \mid \bar{s}, a) := \sum_{s' \in \bar{s}'} P(s' \mid s, a), \quad (4)$$

$$\bar{r}^i(\bar{s}, a) := r^i(s, a), \quad (5)$$

$$\bar{\mu}(\bar{s}) := \sum_{s \in \bar{s}} \mu(s).$$

When M is an MPG satisfying Theorem 5, the corresponding orbit-level potential signal is

$$\bar{\phi}(\bar{s}, a) := \phi(s, a) \quad \text{for any } s \in \bar{s}. \quad (6)$$

Quotient-process notation. For a quotient policy $\bar{\pi}$, write $\bar{V}_{\bar{\pi}}^i$ and $\bar{Q}_{\bar{\pi}}^i$ for agent i 's value and action-value functions in $\bar{M}$, and write $\bar{d}_{\bar{\mu}}^{\bar{\pi}}$ for its discounted state-visitation measure from $\bar{\mu}$. These quantities use the same definitions as in Section 1, with $(M, \mathcal{S}, P, r, \mu)$ replaced by $(\bar{M}, \bar{\mathcal{S}}, \bar{P}, \bar{r}, \bar{\mu})$. Let $\Pi(\bar{M})$ denote the stationary product policies on the quotient and let $\Pi_G(M)$ denote the G -invariant stationary product policies on M .

Lifting quotient policies. Let $\bar{\pi}$ be a stationary product policy on $\bar{M}$. Its lift to M is the product policy $\bar{\pi}^\uparrow$ defined coordinatewise by

$$\bar{\pi}^{\uparrow,i}(a^i | s) := \bar{\pi}^i(a^i | \rho(s)).$$

The lift is G invariant because $\rho(L_g[s]) = \rho(s)$.

Conversely, a G -invariant policy π on M can be projected to $\bar{M}$ by

$$\pi^{\downarrow,i}(a^i | \bar{s}) := \pi^i(a^i | \iota(\bar{s})).$$

Invariance makes this definition independent of the selected representative. Thus quotient policies are in bijection with G -invariant policies on M .

Black-box rate convention. All rate statements below use a target accuracy $\epsilon > 0$. Fix a learning algorithm. A function $T_{\text{MG}}(\epsilon, S; \vartheta)$ is a *uniform MPG certificate* on a class of games if, for every game X in the class,

$$T \geq T_{\text{MG}}(\epsilon, |S_X|; \vartheta) \implies \min_{0 \leq t < T} \text{Nash gap}_X(\pi_t^X) \leq \epsilon. \quad (7)$$

Here $S = |S_X|$ is the number of Markov states and ϑ collects every other hypothesis and numerical bound used by the convergence theorem. Calling the certificate uniform means that the same ϑ is valid for all games being compared.

Theorem 8 (Exact quotient reduction and unrestricted Nash preservation). *Let M be a G -state-symmetric MPG under Theorems 3 and 5. Then the quotient $\bar{M}$ has the following properties.*

1. *The quotient data in Theorem 7 do not depend on the representative $s \in \bar{s}$. The resulting quotient is an MPG with potential $\bar{\phi}$ and total-potential bounds $[\Phi_{\min}, \Phi_{\max}]$. For every quotient policy $\bar{\pi}$, its lift is G invariant and preserves statewise values:*

$$V_{\bar{\pi}^\uparrow}^i(s) = \bar{V}_{\bar{\pi}}^i(\rho(s)).$$

2. *Lifting preserves the unrestricted Nash gap:*

$$\text{Nash gap}_M(\bar{\pi}^\uparrow) = \text{Nash gap}_{\bar{M}}(\bar{\pi}). \quad (8)$$

The maximization on the left still ranges over all local policies on M . Consequently, $\bar{\pi}$ is ϵ Nash in $\bar{M}$ if and only if $\bar{\pi}^\uparrow$ is ϵ Nash in M .

3. *Suppose the uniform certificate above applies to a class containing both M and $\bar{M}$. Run the algorithm on $\bar{M}$ and lift every iterate as $\pi_t := \bar{\pi}_t^\uparrow$. Then*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{MG}}(\epsilon, |\bar{S}|; \vartheta). \quad (9)$$

Thus any uniform rate for the unreduced MPG transfers by substituting $|\bar{S}|$ for $|S|$. If the certificate is nondecreasing in S , this sufficient iteration budget is no larger than the unreduced one. In particular, if a sufficient budget has state dependence

$$B_X(\epsilon) = C(\vartheta) |S_X|^\alpha \epsilon^{-p}, \quad C(\vartheta) > 0, \quad \alpha, p \geq 0,$$

then the full and quotient sufficient budgets satisfy

$$\frac{B_{\text{quot}}(\epsilon)}{B_{\text{full}}(\epsilon)} = \left(\frac{|\bar{S}|}{|S|} \right)^\alpha. \quad (10)$$

This is a comparison of sufficient upper bounds, not realized runtime.

Proof idea. Transition equivariance makes the orbit process Markov, and a Bellman fixed-point argument identifies quotient values with lifted values. Against invariant opponents, an agent's best-response MDP is itself G symmetric. Its optimal action values are constant on state orbits, so fixed tie-breaking gives an invariant optimal response. Quotient deviations therefore attain the same value as unrestricted deviations in the original game. The rate statement then applies the uniform certificate to the quotient iterates and lifts them. Formal proofs are in Sections A to E.

Remark 9 (Scope of the quotient). The construction uses the same joint action a at every state in an orbit. If the group also transforms action labels or permutes agent identities, then equation 4 does not by itself define the appropriate reduced decision problem. Such symmetries require a reduction that also tracks the moved action or agent coordinates. Homogeneous agent relabelings are treated through pointed-state policy sharing in Section 4.

4 HOMOGENEOUS AGENT-PERMUTATION SYMMETRY

State quotients identify states while keeping every decision coordinate fixed. Relabeling homogeneous agents instead moves state, action, reward, and policy coordinates together. The appropriate exact representation is therefore a shared policy on orbits of *pointed states*, where the point records the acting agent. Let S_N denote the symmetric group of permutations of $\mathcal{N} = [N]$. As in Section 1, each local policy is indexed by the full state; the pointed orbit below is its invariant summary relative to a focal agent.

Assumption 10 (Homogeneous agent-permutation structure). Let $\mathcal{C}$ and $\mathcal{B}$ be finite sets, let $\mathcal{S} = \mathcal{C}^N$, and let $\mathcal{A}^i = \mathcal{B}$ for every agent. For $\sigma \in S_N$, define

$$(\sigma \cdot s)^i := s^{\sigma^{-1}(i)}, \quad (\sigma \cdot a)^i := a^{\sigma^{-1}(i)}.$$

Assume that, for every $\sigma \in S_N$, $i \in \mathcal{N}$, $s, s' \in \mathcal{S}$, and $a \in \mathcal{B}^N$,

$$P(\sigma \cdot s' \mid \sigma \cdot s, \sigma \cdot a) = P(s' \mid s, a), \quad (11)$$

$$r^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = r^i(s, a). \quad (12)$$

For the MPG potential and initial distribution, assume

$$\begin{aligned} \phi(\sigma \cdot s, \sigma \cdot a) &= \phi(s, a), \\ \mu(\sigma \cdot s) &= \mu(s). \end{aligned} \quad (13)$$

Let S_N act on a pointed state by $\sigma \cdot (i, s) := (\sigma(i), \sigma \cdot s)$, and define

$$\mathcal{Z} := (\mathcal{N} \times \mathcal{S})/S_N, \quad z_i(s) := [(i, s)]_{S_N}.$$

Thus $z_i(s)$ is the orbit of the state as seen from focal agent i . In the Cartesian model $\mathcal{S} = \mathcal{C}^N$, this orbit has the concrete representation

$$z_i(s) \longleftrightarrow (s^i, \{\{s^k : k \neq i\}\}).$$

Equivalently, $\mathcal{Z} \cong \mathcal{C} \times \mathcal{M}_{N-1}(\mathcal{C})$, where $\mathcal{M}_m(\mathcal{C})$ denotes the multisets of size m drawn from $\mathcal{C}$. Hence

$$|\mathcal{Z}| = |\mathcal{C}| \binom{|\mathcal{C}| + N - 2}{N - 1}.$$

Thus sharing uses $|\mathcal{Z}|$ local policy blocks instead of the $N |\mathcal{C}|^N$ blocks in an unrestricted product-policy table.

Definition 11 (Permutation-invariant and shared policies). A product policy π is *permutation invariant* if

$$\pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s) \quad \text{for every } \sigma \in S_N, i \in \mathcal{N}, s \in \mathcal{S}, \text{ and } b \in \mathcal{B}. \quad (14)$$

A product policy is *pointed-orbit shared* if there is one rule $\psi : \mathcal{Z} \rightarrow \Delta(\mathcal{B})$ such that

$$\pi^i(b \mid s) = \psi(b \mid z_i(s)) \quad \text{for every } i \in \mathcal{N}, s \in \mathcal{S}, \text{ and } b \in \mathcal{B}. \quad (15)$$

Write $\Pi_{S_N}(M)$ for the permutation-invariant product policies.

For any rule $\chi : \mathcal{Z} \rightarrow \Delta(\mathcal{B})$ and focal agent i , write $\chi \circ z_i$ for the local policy

$$(\chi \circ z_i)(b \mid s) := \chi(b \mid z_i(s)).$$

Shared-solver rate convention. Call an MPG X *permutation symmetric* if it has agent set $\mathcal{N} = [N]$, a common action set $\mathcal{B}_X$, a finite state space $\mathcal{S}_X$ equipped with a S_N action, and satisfies the transition, reward, potential, and initial-distribution symmetry identities in Theorem 10. Let S_N act on pointed states by $\sigma \cdot (i, x) := (\sigma(i), \sigma \cdot x)$, and define

$$\begin{aligned} \mathcal{Z}_X &:= (\mathcal{N} \times \mathcal{S}_X)/S_N, & z_i^X(x) &:= [(i, x)]_{S_N}, \\ \text{OrbGap}_X^i(\pi) &:= \max_{\chi : \mathcal{Z}_X \rightarrow \Delta(\mathcal{B}_X)} \left[V_{\chi \circ z_i^X, \pi^{-i}}^i(\mu_X) - V_{\pi}^i(\mu_X) \right]. \end{aligned} \quad (16)$$

Here $(\chi \circ z_i^X)(b \mid x) := \chi(b \mid z_i^X(x))$, and μ_X is the initial distribution of X . A *pointed-orbit shared solver* keeps every iterate in the class $\pi^i(b \mid x) = \psi(b \mid z_i^X(x))$. For a fixed such solver, a function $T_{\text{sh}}(\epsilon, S, D; \vartheta)$ is a *uniform shared certificate* if, for any fixed agent i ,

$$T \geq T_{\text{sh}}(\epsilon, |S_X|, |\mathcal{Z}_X|; \vartheta) \implies \min_{0 \leq t < T} \text{OrbGap}_X^i(\pi_t) \leq \epsilon. \quad (17)$$

Here $S = |S_X|$, $D = |\mathcal{Z}_X|$, and ϑ again collects all common hypotheses and bounds. For the present game, $\mathcal{Z}_M = \mathcal{Z}$ and $z_i^M = z_i$.

Theorem 12 (Exact agent-permutation sharing and unrestricted best responses). *Let M be an MPG satisfying Theorem 10. Then:*

1. *The concrete pair $(s^i, \{\{s^k : k \neq i\}\})$ uniquely represents $z_i(s)$. Permutation-invariant product policies are exactly pointed-orbit shared policies, so equation 15 is a bijection between rules on $\mathcal{Z}$ and policies in $\Pi_{S_N}(M)$.*
2. *For every $\pi \in \Pi_{S_N}(M)$ and any fixed agent i , against the invariant opponents π^{-i} , an unrestricted optimal stationary response can be chosen in shared form:*

$$\max_{\nu^i: S \rightarrow \Delta(\mathcal{B})} V_{\nu^i, \pi^{-i}}^i(\mu) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} V_{\chi \circ z_i, \pi^{-i}}^i(\mu). \quad (18)$$

The best-response gains are equal across agents. Hence, for every i ,

$$\text{Nash gap}_M(\pi) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ z_i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right]. \quad (19)$$

This is the ordinary Nash gap from Theorem 2, whose definition allows every stationary local deviation.

3. *If a pointed-orbit shared solver has the uniform certificate above on a class containing M , then its iterates on M satisfy*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{sh}}(\epsilon, |\mathcal{C}|^N, |\mathcal{Z}|; \vartheta). \quad (20)$$

For a bound comparison, additionally suppose that the rate analysis is orbit compatible: an untied counterpart controls the ordinary Nash gap with the same certificate form when $D = N|S_X|$, the shared implementation controls equation 16 with $D = |\mathcal{Z}_X|$, and both have sufficient budgets of the form

$$B_{\text{mode}}(\epsilon) = C_{\text{mode}}(\vartheta) S^\alpha D_{\text{mode}}^\delta \epsilon^{-p}, \quad \alpha, \delta, p \geq 0,$$

where $D_{\text{full}} = N|S_X|$ and $D_{\text{perm}} = |\mathcal{Z}_X|$. Assume the positive prefactors satisfy $C_{\text{perm}}(\vartheta) \leq C_{\text{full}}(\vartheta)$. Then

$$\frac{B_{\text{perm}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq \left(\frac{|\mathcal{Z}|}{N|\mathcal{C}|^N} \right)^\delta. \quad (21)$$

Exact symmetry gives the transfer in equation 20; it does not by itself imply orbit compatibility. In particular, constants affected by orbit multiplicities must be absorbed into C_{mode} , and the prefactor inequality is an additional assumption. The ratio compares sufficient iteration bounds, not realized runtime.

Proof idea. A permutation-invariant local policy is constant on the orbit of the pointed state (i, s) , which gives the shared-rule representation. For a fixed agent i , the best-response MDP is symmetric under the permutations that fix i ; their state orbits are exactly the fibers of z_i . A symmetric optimal response therefore exists, while relabeling agents shows that every agent has the same best-response gain. This converts the shared solver’s pointed-orbit gap into the ordinary Nash gap. Formal proofs and supporting equivariance identities are in Sections F to I.

Remark 13 (Sharing rather than a state quotient). The orbit $z_i(s)$ depends on a focal agent. Thus Theorem 12 does not replace the original game by a Markov game on $\mathcal{Z}$ and does not require a unilateral deviator to preserve global permutation invariance. It says that invariant product policies have one shared representation and that, at such a policy, a best response for agent i can be chosen invariant under permutations that fix i .

Remark 14 (Automatic potential invariance). In a fully cooperative game with $r^i \equiv r$, the choice $\phi = r$ is permutation invariant whenever the common reward is. More generally, if $\phi(s, a) = N^{-1} \sum_i r^i(s, a)$ is a valid MPG potential, then equation 12 makes it permutation invariant by reindexing the sum over agents.

5 COMBINING STATE QUOTIENTS AND AGENT-PERMUTATION SHARING

The two reductions can be composed when the state-only symmetry is compatible with agent relabeling. Commutativity is the condition that makes agent permutations well defined after passing to state orbits.

Assumption 15 (Commuting state and agent symmetries). Suppose M satisfies both the G -state-symmetry conditions in Theorems 4 and 5 and the homogeneous agent-permutation conditions in Theorem 10. Assume also that

$$L_g[\sigma \cdot s] = \sigma \cdot L_g[s] \quad \text{for every } g \in G, \sigma \in S_N, \text{ and } s \in \mathcal{S}. \quad (22)$$

Under equation 22, $G \times S_N$ acts on pointed states by

$$(g, \sigma) \cdot (i, s) := (\sigma(i), L_g[\sigma \cdot s]).$$

Define the combined pointed-orbit space and map by

$$\mathcal{W} := (\mathcal{N} \times \mathcal{S}) / (G \times S_N), \quad w_i(s) := [(i, s)]_{G \times S_N}. \quad (23)$$

For $\chi : \mathcal{W} \rightarrow \Delta(\mathcal{B})$, write $(\chi \circ w_i)(b \mid s) := \chi(b \mid w_i(s))$. A product policy is *jointly invariant* if it is both G invariant in the sense of Theorem 6 and permutation invariant in the sense of Theorem 11.

Theorem 16 (Commuting reductions and double transfer). *Let M be an MPG under Theorems 3 and 15. Then:*

1. *Agent permutations descend to the state quotient through*

$$\sigma \cdot \bar{s} := \rho(\sigma \cdot s), \quad s \in \bar{s},$$

and $\bar{M}$ is permutation symmetric under this action. Moreover,

$$\mathcal{W} \cong (\mathcal{N} \times (\mathcal{S}/G)) / S_N \cong \mathcal{Z}/G, \quad (24)$$

where the induced action on $\mathcal{Z}$ is $g \cdot [(i, s)]_{S_N} := [(i, L_g[s])]_{S_N}$.

2. *Jointly invariant policies are exactly the policies represented by one rule $\psi : \mathcal{W} \rightarrow \Delta(\mathcal{B})$ through $\pi^i(b \mid s) = \psi(b \mid w_i(s))$. At every such policy and for every fixed agent i ,*

$$\text{Nash gap}_M(\pi) = \max_{\chi : \mathcal{W} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ w_i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) \right]. \quad (25)$$

Thus the maximization still represents the unrestricted Nash gap.

3. *Let a pointed-orbit shared solver have the uniform certificate $T_{\text{sh}}(\epsilon, S, D; \vartheta)$ from equation 17 on a class containing $\bar{M}$. Run it on $\bar{M}$, whose pointed-orbit space is $\mathcal{W}$ by equation 24, and lift its iterates to M . Then*

$$\min_{0 \leq t < T} \text{Nash gap}_M(\pi_t) \leq \epsilon \quad \text{whenever} \quad T \geq T_{\text{sh}}(\epsilon, |\bar{\mathcal{S}}|, |\mathcal{W}|; \vartheta). \quad (26)$$

Suppose additionally that the orbit-compatible untied/shared certificate from Theorem 12 applies to both M and $\bar{M}$, with common exponents $\alpha, \delta, p \geq 0$. Write the four sufficient budgets as

$$B_{\text{mode}}(\epsilon) = C_{\text{mode}}(\vartheta) S_{\text{mode}}^{\alpha} D_{\text{mode}}^{\delta} \epsilon^{-p},$$

where the state and policy-block counts are

mode	full	perm	quot	both
S_{mode}	$ \mathcal{S} $	$ \mathcal{S} $	$ \bar{\mathcal{S}} $	$ \bar{\mathcal{S}} $
D_{mode}	$N \mathcal{S} $	$ \mathcal{Z} $	$N \bar{\mathcal{S}} $	$ \mathcal{W} $

Here “perm” means permutation sharing only, “quot” means the untied state quotient only, and “both” means their composition. Assume the positive prefactors satisfy

$$C_{\text{both}}(\vartheta) \leq \min \{C_{\text{full}}(\vartheta), C_{\text{perm}}(\vartheta), C_{\text{quot}}(\vartheta)\}.$$

The sufficient bounds then satisfy

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq \left(\frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|} \right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|} \right)^\delta \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|} \right)^\delta, \quad (27)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} \leq \left(\frac{|\bar{\mathcal{S}}|}{|\mathcal{S}|} \right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|} \right)^\delta \leq 1. \quad (28)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{quot}}(\epsilon)} \leq \left(\frac{|\mathcal{W}|}{N|\bar{\mathcal{S}}|} \right)^\delta \leq 1. \quad (29)$$

Thus the first bound separates the permutation-only block reduction from the two additional reductions obtained by quotienting the Markov states and the shared policy blocks; the latter two bounds show that adding either reduction to the other is non-worsening under the stated certificate and prefactor assumptions. The required prefactor inequalities are assumptions on the convergence analysis, not consequences of symmetry. If G acts freely on $\mathcal{S}$ and its induced action on $\mathcal{Z}$ is also free, both $|\bar{\mathcal{S}}|/|\mathcal{S}|$ and $|\mathcal{W}|/|\mathcal{Z}|$ equal $|G|^{-1}$, so

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} \leq |G|^{-(\alpha+\delta)}, \quad (30)$$

$$\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} \leq |G|^{-(\alpha+\delta)} \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|} \right)^\delta. \quad (31)$$

Proof idea. Commutativity lets agent permutations act on G -orbits, so quotienting states and then taking pointed-state orbits identifies the same classes as the direct $G \times S_N$ action. At a jointly invariant policy, a focal agent's best-response MDP is symmetric under G and the agent permutations that fix that agent. The resulting invariant optimal response is constant on the fibers of w_i . Running the shared solver on the quotient and lifting its iterates then combines the two exact Nash-gap identities. The detailed orbit and rate calculations are given in Section J.

A WELL-POSEDNESS OF THE QUOTIENT MARKOV GAME

Lemma 17 (Well-posed quotient data). *Let M be a G -state-symmetric MPG satisfying Theorem 5. Then the quotient data in Theorem 7 are independent of the representative $s \in \bar{s}$. Moreover, $\bar{P}(\cdot \mid \bar{s}, a) \in \Delta(\bar{\mathcal{S}})$ for every $(\bar{s}, a)$ and $\bar{\mu} \in \Delta(\bar{\mathcal{S}})$.*

Proof. Let $s, \tilde{s} \in \bar{s}$. There is a $g \in G$ such that $\tilde{s} = L_g[s]$. For every next-state orbit $\bar{s}'$, the map $u \mapsto L_g[u]$ is a bijection from $\bar{s}'$ to itself. Reindexing the orbit sum and using transition equivariance,

$$\begin{aligned} \sum_{u \in \bar{s}'} P(u \mid \tilde{s}, a) &= \sum_{u \in \bar{s}'} P(L_g[u] \mid L_g[s], a) \\ &= \sum_{u \in \bar{s}'} P(u \mid s, a). \end{aligned}$$

Thus equation 4 is independent of the representative. Reward and potential invariance similarly give

$$r^i(\tilde{s}, a) = r^i(s, a), \quad \phi(\tilde{s}, a) = \phi(s, a),$$

so equation 5 and equation 6 are also representative-independent. The definition of $\bar{\mu}$ already sums over the full orbit and contains no representative choice.

The state orbits partition $\mathcal{S}$. Hence, for any representative $s \in \bar{s}$,

$$\begin{aligned} \sum_{\bar{s}' \in \bar{\mathcal{S}}} \bar{P}(\bar{s}' \mid \bar{s}, a) &= \sum_{\bar{s}' \in \bar{\mathcal{S}}} \sum_{s' \in \bar{s}'} P(s' \mid s, a) \\ &= \sum_{s' \in \mathcal{S}} P(s' \mid s, a) = 1. \end{aligned}$$

Every term is nonnegative, so $\bar{P}(\cdot \mid \bar{s}, a) \in \Delta(\bar{\mathcal{S}})$. Likewise,

$$\sum_{\bar{s} \in \bar{\mathcal{S}}} \bar{\mu}(\bar{s}) = \sum_{\bar{s} \in \bar{\mathcal{S}}} \sum_{s \in \bar{s}} \mu(s) = \sum_{s \in \mathcal{S}} \mu(s) = 1,$$

and $\bar{\mu}$ is nonnegative. Thus $\bar{\mu} \in \Delta(\bar{\mathcal{S}})$. $\square$

B POLICY CORRESPONDENCE AND VALUE EQUIVALENCE

Lemma 18 (Policy correspondence and value equivalence). *Let M be a G -state-symmetric Markov game and let $\bar{M}$ be its quotient. Then the lift and projection satisfy:*

(a) *The maps $\bar{\pi} \mapsto \bar{\pi}^\uparrow$ and $\pi \mapsto \pi^\downarrow$ are mutually inverse bijections between $\Pi(\bar{M})$ and $\Pi_G(M)$.*

(b) *For every $\bar{\pi} \in \Pi(\bar{M})$, agent i , state s , and joint action a ,*

$$V_{\bar{\pi}^\uparrow}^i(s) = \bar{V}_{\bar{\pi}}^i(\rho(s)), \quad Q_{\bar{\pi}^\uparrow}^i(s, a) = \bar{Q}_{\bar{\pi}}^i(\rho(s), a). \quad (32)$$

Consequently, $V_{\bar{\pi}^\uparrow}^i(\mu) = \bar{V}_{\bar{\pi}}^i(\bar{\mu})$.

(c) *For every $\bar{\pi} \in \Pi(\bar{M})$ and $\bar{s} \in \bar{\mathcal{S}}$,*

$$\bar{d}_{\bar{\mu}}^{\bar{\pi}}(\bar{s}) = \sum_{s \in \bar{s}} d_{\mu}^{\bar{\pi}^\uparrow}(s). \quad (33)$$

Proof. For every $g \in G$, $\rho(L_g[s]) = \rho(s)$. Therefore

$$\bar{\pi}^{\uparrow, i}(a^i \mid L_g[s]) = \bar{\pi}^i(a^i \mid \rho(L_g[s])) = \bar{\pi}^i(a^i \mid \rho(s)) = \bar{\pi}^{\uparrow, i}(a^i \mid s),$$

so every lift is G invariant. The projection is well defined because an invariant policy is constant on each orbit. Direct substitution gives

$$(\bar{\pi}^\uparrow)^\downarrow = \bar{\pi}, \quad (\pi^\downarrow)^\uparrow = \pi \quad \text{for every } \pi \in \Pi_G(M).$$

This proves part (a).

Fix a quotient policy $\bar{\pi}$ and write $\pi = \bar{\pi}^\dagger$. Define

$$\tilde{V}^i(s) := \bar{V}_{\bar{\pi}}^i(\rho(s)).$$

Using $\pi(a | s) = \bar{\pi}(a | \rho(s))$ and the quotient definitions,

$$\begin{aligned} \tilde{V}^i(s) &= \sum_a \bar{\pi}(a | \rho(s)) \left[\bar{r}^i(\rho(s), a) + \gamma \sum_{\bar{s}'} \bar{P}(\bar{s}' | \rho(s), a) \bar{V}_{\bar{\pi}}^i(\bar{s}') \right] \\ &= \sum_a \pi(a | s) \left[r^i(s, a) + \gamma \sum_{\bar{s}'} \sum_{s' \in \bar{s}'} P(s' | s, a) \tilde{V}^i(s') \right] \\ &= \sum_a \pi(a | s) \left[r^i(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) \tilde{V}^i(s') \right]. \end{aligned}$$

Thus $\tilde{V}^i$ is a fixed point of the Bellman evaluation operator for π on M . That operator is a γ -contraction, so its fixed point is unique and $\tilde{V}^i = V_{\pi}^i$. Substituting this value identity into the one-step action-value equations gives

$$\begin{aligned} Q_{\pi}^i(s, a) &= r^i(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\pi}^i(s') \\ &= \bar{r}^i(\rho(s), a) + \gamma \sum_{\bar{s}'} \bar{P}(\bar{s}' | \rho(s), a) \bar{V}_{\bar{\pi}}^i(\bar{s}') \\ &= \bar{Q}_{\bar{\pi}}^i(\rho(s), a). \end{aligned}$$

This proves equation 32. Averaging its value identity over $s \sim \mu$ and grouping by state orbit yields

$$\begin{aligned} V_{\pi^\dagger}^i(\mu) &= \sum_{\bar{s}} \sum_{s \in \bar{s}} \mu(s) \bar{V}_{\bar{\pi}}^i(\bar{s}) \\ &= \sum_{\bar{s}} \bar{\mu}(\bar{s}) \bar{V}_{\bar{\pi}}^i(\bar{s}) = \bar{V}_{\bar{\pi}}^i(\bar{\mu}). \end{aligned}$$

For part (c), the state-to-state kernels induced by π and $\bar{\pi}$ satisfy

$$\sum_{s' \in \bar{s}'} P^\pi(s' | s) = \bar{P}^{\bar{\pi}}(\bar{s}' | \rho(s)). \quad (34)$$

Indeed, this follows by summing $P^\pi(s' | s) = \sum_a \pi(a | s) P(s' | s, a)$ over $\bar{s}'$ and applying equation 4. Let $p_t^\pi(s' | s_0)$ and $\bar{p}_t^{\bar{\pi}}(\bar{s}' | \bar{s}_0)$ denote the corresponding t -step transition probabilities. We claim that

$$\sum_{s' \in \bar{s}'} p_t^\pi(s' | s_0) = \bar{p}_t^{\bar{\pi}}(\bar{s}' | \rho(s_0)) \quad \text{for every } t \geq 0. \quad (35)$$

At $t = 0$, both sides are the indicator of $\rho(s_0) = \bar{s}'$. If the identity holds at time t , then grouping the one-step recursion for p_{t+1}^π by the orbit $\bar{u}$ of the intermediate state and using equation 34 gives

$$\begin{aligned} \sum_{s' \in \bar{s}'} p_{t+1}^\pi(s' | s_0) &= \sum_{\bar{u} \in \bar{\mathcal{S}}} \left(\sum_{u \in \bar{u}} p_t^\pi(u | s_0) \right) \bar{P}^{\bar{\pi}}(\bar{s}' | \bar{u}) \\ &= \sum_{\bar{u} \in \bar{\mathcal{S}}} \bar{p}_t^{\bar{\pi}}(\bar{u} | \rho(s_0)) \bar{P}^{\bar{\pi}}(\bar{s}' | \bar{u}) \\ &= \bar{p}_{t+1}^{\bar{\pi}}(\bar{s}' | \rho(s_0)). \end{aligned}$$

Thus equation 35 follows by induction. Finally,

$$\begin{aligned}\bar{d}_\mu^\pi(\bar{s}') &= \sum_{t=0}^{\infty} \gamma^t \sum_{\bar{s}_0} \bar{\mu}(\bar{s}_0) \bar{p}_t^\pi(\bar{s}' | \bar{s}_0) \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{s_0 \in \mathcal{S}} \mu(s_0) \sum_{s' \in \bar{s}'} p_t^\pi(s' | s_0) \\ &= \sum_{s' \in \bar{s}'} d_\mu^\pi(s'),\end{aligned}$$

where the second equality uses $\bar{\mu}(\bar{s}_0) = \sum_{s_0 \in \bar{s}_0} \mu(s_0)$ and equation 35. This is equation 33. $\square$

C THE QUOTIENT IS A MARKOV POTENTIAL GAME

Lemma 19 (Inherited MPG structure and regularity). *Let M be a G -state-symmetric MPG satisfying Theorems 3 and 5. Then its quotient has the following properties:*

- (a) *It is an MPG with potential $\bar{\phi}$ and total-potential bounds $[\Phi_{\min}, \Phi_{\max}]$.*
- (b) *Every quotient policy has positive discounted visitation on every quotient state.*
- (c) *Its mismatch coefficient*

$$\kappa_G := \max_{\bar{\pi}, \bar{\pi}' \in \Pi(\bar{M})} \left\| \bar{d}_\mu^\pi / \bar{d}_\mu^{\pi'} \right\|_\infty$$

satisfies $\kappa_G \leq \kappa$.

Proof. Let $\bar{\Phi}_\pi$ denote the discounted total potential generated by $\bar{\phi}$ in $\bar{M}$. The Bellman argument in Theorem 18(b), with ϕ and $\bar{\phi}$ in place of r^i and $\bar{r}^i$, gives

$$\Phi_{\bar{\pi}^\uparrow}(s) = \bar{\Phi}_\pi(\rho(s)). \quad (36)$$

Fix an agent i , a quotient policy $\bar{\pi}$, and an alternative local policy $\bar{\nu}^i$. Set $\tilde{\pi} := (\bar{\nu}^i, \bar{\pi}^{-i})$. Lifting acts coordinatewise, so $\tilde{\pi}^\uparrow$ and $\bar{\pi}^\uparrow$ differ only in agent i 's policy. At any representative $s \in \bar{s}$, value equivalence, the MPG identity of M , and equation 36 give

$$\begin{aligned}\bar{V}_{\tilde{\pi}}^i(\bar{s}) - \bar{V}_{\bar{\pi}}^i(\bar{s}) &= V_{\tilde{\pi}^\uparrow}^i(s) - V_{\bar{\pi}^\uparrow}^i(s) \\ &= \Phi_{\tilde{\pi}^\uparrow}(s) - \Phi_{\bar{\pi}^\uparrow}(s) \\ &= \bar{\Phi}_{\tilde{\pi}}(\bar{s}) - \bar{\Phi}_{\bar{\pi}}(\bar{s}).\end{aligned}$$

Hence $\bar{\phi}$ is a potential for $\bar{M}$. Moreover,

$$\bar{\Phi}_{\bar{\pi}}(\bar{s}) = \Phi_{\bar{\pi}^\uparrow}(s) \in [\Phi_{\min}, \Phi_{\max}],$$

which proves part (a).

By equation 33,

$$\bar{d}_\mu^\pi(\bar{s}) = \sum_{s \in \bar{s}} d_\mu^{\bar{\pi}^\uparrow}(s).$$

Each summand is positive by Theorem 3, proving part (b).

For quotient policies $\bar{\pi}, \bar{\pi}'$ and a quotient state $\bar{s}$,

$$\begin{aligned}\frac{\bar{d}_\mu^\pi(\bar{s})}{\bar{d}_\mu^{\pi'}(\bar{s})} &= \frac{\sum_{s \in \bar{s}} d_\mu^{\bar{\pi}^\uparrow}(s)}{\sum_{s \in \bar{s}} d_\mu^{\bar{\pi}'^\uparrow}(s)} \\ &\leq \max_{s \in \bar{s}} \frac{d_\mu^{\bar{\pi}^\uparrow}(s)}{d_\mu^{\bar{\pi}'^\uparrow}(s)} \leq \kappa.\end{aligned}$$

The first inequality is the mediant inequality for positive numbers; the second is the definition of κ . Maximizing over quotient states and policies proves $\kappa_G \leq \kappa$. $\square$

D NASH-GAP PRESERVATION

The key point is that, against invariant opponents, an agent has an invariant optimal best response even though the original Nash gap allows all local policies.

Lemma 20 (Invariant optimal best response). *Let M be a G -state-symmetric Markov game. Fix an agent i and a G -invariant product policy π . The best-response MDP faced by agent i when the other agents play π^{-i} is G -state-symmetric and has a G -invariant optimal stationary policy. Consequently,*

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) = \max_{\tilde{\pi}^i \text{ } G\text{-invariant}} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu). \quad (37)$$

Proof. The best-response MDP has state space $\mathcal{S}$, action space $\mathcal{A}^i$, discount γ , initial distribution μ , and data

$$\begin{aligned} P_{\pi^{-i}}^i(s' \mid s, a^i) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) P(s' \mid s, a^i, a^{-i}), \\ r_{\pi^{-i}}^i(s, a^i) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) r^i(s, a^i, a^{-i}). \end{aligned}$$

Since π^{-i} is invariant and the group leaves every action label fixed,

$$\begin{aligned} P_{\pi^{-i}}^i(L_g[s'] \mid L_g[s], a^i) &= \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid L_g[s]) P(L_g[s'] \mid L_g[s], a^i, a^{-i}) \\ &= \sum_{a^{-i}} \pi^{-i}(a^{-i} \mid s) P(s' \mid s, a^i, a^{-i}) \\ &= P_{\pi^{-i}}^i(s' \mid s, a^i), \end{aligned}$$

where the second equality uses policy invariance and equation 3. The same calculation with rewards gives

$$r_{\pi^{-i}}^i(L_g[s], a^i) = r_{\pi^{-i}}^i(s, a^i).$$

Thus the best-response MDP is G -state-symmetric.

Let $\mathcal{T}$ be its Bellman optimality operator. If a function V is G invariant, then transition and reward symmetry imply

$$(\mathcal{T}V)(L_g[s]) = (\mathcal{T}V)(s).$$

Starting value iteration from the zero function therefore produces only invariant functions. Because $\mathcal{T}$ is a γ -contraction, these iterates converge to the unique optimal value V^* , which is invariant. Its optimal action-value function consequently satisfies

$$Q^*(L_g[s], a^i) = Q^*(s, a^i).$$

Choose a greedy deterministic policy and break ties using one fixed total order on $\mathcal{A}^i$ at every state. The action-value vectors are identical within each orbit, so the selected greedy action is also identical within each orbit. This gives a G -invariant optimal stationary policy. Since this policy attains the unrestricted best-response value, equation 37 follows. $\square$

Lemma 21 (Nash-gap preservation). *Let M be a G -state-symmetric Markov game. Every quotient policy $\bar{\pi}$ satisfies*

$$\text{Nash gap}_M(\bar{\pi}^\uparrow) = \text{Nash gap}_{\bar{M}}(\bar{\pi}).$$

Consequently, $\bar{\pi}$ is $\in \text{Nash}$ in $\bar{M}$ if and only if $\bar{\pi}^\uparrow$ is $\in \text{Nash}$ in M .

Proof. Write $\pi = \bar{\pi}^\uparrow$. Fix an agent i . Since π^{-i} is invariant, Theorem 20 allows the original best-response optimization to be restricted to invariant local deviations:

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) = \max_{\tilde{\pi}^i \text{ } G\text{-invariant}} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu).$$

By Theorem 18(a), projection is a bijection between invariant local policies on M and local policies on $\overline{M}$. Concretely, for an invariant deviation $\tilde{\pi}^i$, define

$$\bar{\nu}^i(a^i | \bar{s}) := \tilde{\pi}^i(a^i | \iota(\bar{s})).$$

Then $(\tilde{\pi}^i, \pi^{-i}) = (\bar{\nu}^i, \bar{\pi}^{-i})^\uparrow$. Value equivalence gives

$$\begin{aligned} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) &= \bar{V}_{\bar{\nu}^i, \bar{\pi}^{-i}}^i(\bar{\mu}), \\ V_{\pi}^i(\mu) &= \bar{V}_{\bar{\pi}}^i(\bar{\mu}). \end{aligned}$$

Therefore,

$$\max_{\tilde{\pi}^i} V_{\tilde{\pi}^i, \pi^{-i}}^i(\mu) - V_{\pi}^i(\mu) = \max_{\bar{\nu}^i} \bar{V}_{\bar{\nu}^i, \bar{\pi}^{-i}}^i(\bar{\mu}) - \bar{V}_{\bar{\pi}}^i(\bar{\mu}).$$

The left and right sides are respectively agent i 's contributions to the Nash gaps in M and $\overline{M}$. Maximizing over i proves the identity in equation 8. $\square$

E BLACK-BOX RATE TRANSFER THROUGH THE QUOTIENT

Proof of Theorem 8, part 3. By hypothesis, the uniform MPG certificate applies to $\overline{M}$. Its state count is $|\overline{\mathcal{S}}|$, while the certificate parameters collected in ϑ are valid for both games. Applying equation 7 to $\overline{M}$ gives

$$\min_{0 \leq t < T} \text{Nash gap}_{\overline{M}}(\bar{\pi}_t) \leq \epsilon$$

whenever $T \geq T_{\text{MG}}(\epsilon, |\overline{\mathcal{S}}|; \vartheta)$. For every iterate, equation 8 gives

$$\text{Nash gap}_M(\pi_t) = \text{Nash gap}_M(\pi_t^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

This proves equation 9.

For the displayed monomial certificate, applying the same sufficient-bound formula to M and $\overline{M}$ gives

$$\begin{aligned} B_{\text{full}}(\epsilon) &= C(\vartheta) |\mathcal{S}|^\alpha \epsilon^{-p}, \\ B_{\text{quot}}(\epsilon) &= C(\vartheta) |\overline{\mathcal{S}}|^\alpha \epsilon^{-p}. \end{aligned}$$

Taking their ratio proves equation 10. The argument uses only a convergence theorem for MPGs and does not invoke a single-agent convergence result.

If a particular external rate theorem also depends monotonically on visitation regularity, that theorem may additionally use Theorem 19(b) and (c). In particular, the original mismatch coefficient κ is a common valid upper bound because the quotient coefficient is at most κ . These secondary refinements are not needed for the state-count transfer stated in the main theorem. $\square$

F PERMUTATION INVARIANCE AND SHARED POINTED-STATE POLICIES

Lemma 22 (Pointed-state orbits and shared policies). *For pointed states (i, s) and (j, s') , the equality $z_i(s) = z_j(s')$ holds if and only if*

$$s^i = s'^j, \quad \{\{s^k : k \neq i\}\} = \{\{s'^k : k \neq j\}\}.$$

Moreover, a product policy is permutation invariant if and only if it is pointed-orbit shared.

Proof. If $z_i(s) = z_j(s')$, then, by definition of the orbit, some $\sigma \in S_N$ satisfies $j = \sigma(i)$ and $s' = \sigma \cdot s$. Then

$$s'^j = (\sigma \cdot s)^{\sigma(i)} = s^i,$$

and the coordinates of s' other than j are a reordering of the coordinates of s other than i . Hence the two concrete descriptors agree.

Conversely, suppose the two concrete descriptors agree. Set $\sigma(i) = j$. Equality of their multisets gives a bijection

$$\sigma : \mathcal{N} \setminus \{i\} \longrightarrow \mathcal{N} \setminus \{j\}$$

such that $s'^{\sigma(k)} = s^k$ for every $k \neq i$. Together with $s'^j = s^i$, this defines a permutation of $\mathcal{N}$ satisfying

$$(\sigma \cdot s)^{\sigma(k)} = s^k = s'^{\sigma(k)} \quad \text{for every } k \in \mathcal{N}.$$

Thus $\sigma \cdot s = s'$ and $\sigma \cdot (i, s) = (j, s')$. This proves the orbit characterization.

Now let π be permutation invariant. If $z_i(s) = z_j(s')$, choose σ as above. Then equation 14 gives

$$\pi^j(b \mid s') = \pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s).$$

Consequently,

$$\psi(b \mid z) := \pi^i(b \mid s) \quad \text{for any } (i, s) \text{ such that } z_i(s) = z$$

is well defined and gives equation 15.

Conversely, suppose equation 15 holds. For every σ ,

$$z_{\sigma(i)}(\sigma \cdot s) = z_i(s),$$

because the two pointed states are in the same orbit. Therefore

$$\pi^{\sigma(i)}(b \mid \sigma \cdot s) = \psi(b \mid z_{\sigma(i)}(\sigma \cdot s)) = \psi(b \mid z_i(s)) = \pi^i(b \mid s),$$

so π is permutation invariant.

Every $z \in \mathcal{Z}$ is the orbit of some pointed state. It follows that the shared rule inducing an invariant policy is unique, so the correspondence is a bijection. Finally, per-agent invariance implies the joint identity

$$\pi(\sigma \cdot a \mid \sigma \cdot s) = \pi(a \mid s), \tag{38}$$

because

$$\pi(\sigma \cdot a \mid \sigma \cdot s) = \prod_{i \in \mathcal{N}} \pi^{\sigma(i)}(a^i \mid \sigma \cdot s) = \prod_{i \in \mathcal{N}} \pi^i(a^i \mid s).$$

□

G VALUE, ADVANTAGE, VISITATION, AND POTENTIAL EQUIVARIANCE

Lemma 23 (Strategic equivariance under agent relabeling). *Under Theorem 10, every $\pi \in \Pi_{S_N}(M)$ satisfies, for every $\sigma \in S_N$, $i \in \mathcal{N}$, $s \in \mathcal{S}$, $a \in \mathcal{B}^N$, and $b \in \mathcal{B}$,*

$$\begin{aligned} V_{\pi}^{\sigma(i)}(\sigma \cdot s) &= V_{\pi}^i(s), & Q_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) &= Q_{\pi}^i(s, a), \\ A_{\pi}^{\sigma(i)}(\sigma \cdot s, b) &= A_{\pi}^i(s, b), & d_{\mu}^{\pi}(\sigma \cdot s) &= d_{\mu}^{\pi}(s), \\ \Phi_{\pi}(\sigma \cdot s) &= \Phi_{\pi}(s). \end{aligned} \tag{39}$$

In particular, there is a well-defined function $\hat{A}_{\pi}$ such that

$$A_{\pi}^i(s, b) = \hat{A}_{\pi}(z_i(s), b). \tag{40}$$

Proof. First consider the state kernel induced by π :

$$P^{\pi}(s' \mid s) := \sum_{a \in \mathcal{B}^N} \pi(a \mid s) P(s' \mid s, a).$$

Reindexing the action sum by $a' = \sigma \cdot a$ and using equation 38 and equation 11 gives

$$\begin{aligned} P^{\pi}(\sigma \cdot s' \mid \sigma \cdot s) &= \sum_{a \in \mathcal{B}^N} \pi(\sigma \cdot a \mid \sigma \cdot s) P(\sigma \cdot s' \mid \sigma \cdot s, \sigma \cdot a) \\ &= P^{\pi}(s' \mid s). \end{aligned} \tag{41}$$

The map

$$(s_t, a_t)_{t \geq 0} \mapsto (\sigma \cdot s_t, \sigma \cdot a_t)_{t \geq 0}$$

is therefore a probability-preserving bijection between trajectories under π started from s and trajectories under π started from $\sigma \cdot s$. By equation 12, the reward of agent $\sigma(i)$ along the transformed trajectory equals the reward of agent i along the original trajectory at every time. It follows that

$$V_{\pi}^{\sigma(i)}(\sigma \cdot s) = V_{\pi}^i(s).$$

Applying the same reindexing to the one-step action-value equations gives

$$Q_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = Q_{\pi}^i(s, a).$$

The same trajectory argument with the invariant signal ϕ in place of r^i proves

$$\Phi_{\pi}(\sigma \cdot s) = \Phi_{\pi}(s).$$

For completeness, let $p_t^{\pi}(s' | s)$ be the t -step transition probability under P^{π} . Equation (41) implies by induction that

$$p_t^{\pi}(\sigma \cdot s' | \sigma \cdot s) = p_t^{\pi}(s' | s) \quad \text{for every } t \geq 0.$$

Using equation 13 and reindexing the initial state as $s_0 = \sigma \cdot u$,

$$\begin{aligned} d_{\mu}^{\pi}(\sigma \cdot s') &= \sum_{t=0}^{\infty} \gamma^t \sum_{s_0 \in \mathcal{S}} \mu(s_0) p_t^{\pi}(\sigma \cdot s' | s_0) \\ &= \sum_{t=0}^{\infty} \gamma^t \sum_{u \in \mathcal{S}} \mu(\sigma \cdot u) p_t^{\pi}(\sigma \cdot s' | \sigma \cdot u) \\ &= d_{\mu}^{\pi}(s'). \end{aligned}$$

Subtracting the value identity from the action-value identity gives joint advantage equivariance:

$$A_{\pi}^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = A_{\pi}^i(s, a). \quad (42)$$

Fix $b \in \mathcal{B}$ and reindex the other agents' actions by $a' = \sigma \cdot a$ with $a^i = b$. Since $(\sigma \cdot a)^{\sigma(i)} = b$, product-policy invariance gives

$$\pi^{-\sigma(i)}\left((\sigma \cdot a)^{-\sigma(i)} \mid \sigma \cdot s\right) = \pi^{-i}(a^{-i} \mid s).$$

Applying this identity and equation 42 in the definition equation 1 yields

$$A_{\pi}^{\sigma(i)}(\sigma \cdot s, b) = A_{\pi}^i(s, b).$$

Finally, if $z_i(s) = z_j(s')$, then Theorem 22 supplies a σ with $j = \sigma(i)$ and $s' = \sigma \cdot s$. The local-advantage identity therefore gives

$$A_{\pi}^j(s', b) = A_{\pi}^i(s, b).$$

Thus defining $\hat{A}_{\pi}(z, b)$ using any pointed state with descriptor z is independent of the choice and proves equation 40. $\square$

H POINTED-ORBIT BEST RESPONSES AND THE UNRESTRICTED NASH GAP

Lemma 24 (Pointed-orbit optimal best response). *Under Theorem 10, let $\pi \in \Pi_{\mathcal{S}_N}(M)$. For every agent i , the best-response identity equation 18 holds. Moreover, all agents have the same best-response gain, and hence equation 19 holds.*

Proof. Fix an agent i and let

$$H_i := \{\tau \in S_N : \tau(i) = i\}$$

be its stabilizer subgroup. Against π^{-i} , agent i faces the finite discounted MDP with state space $\mathcal{S}$, action space $\mathcal{B}$, and

$$\begin{aligned} P_{\pi^{-i}}^i(s' | s, b) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) P(s' | s, b, a^{-i}), \\ r_{\pi^{-i}}^i(s, b) &:= \sum_{a^{-i}} \pi^{-i}(a^{-i} | s) r^i(s, b, a^{-i}). \end{aligned}$$

For $\tau \in H_i$, reindex the sum over a^{-i} by the permutation of the other agents' action coordinates induced by τ . Because $\tau(i) = i$, the focal action b is unchanged. Permutation invariance of π gives

$$\pi^{-i}((\tau \cdot a)^{-i} | \tau \cdot s) = \pi^{-i}(a^{-i} | s) \quad \text{when } a^i = b.$$

Using equation 11 and equation 12, with $\tau(i) = i$, therefore gives

$$P_{\pi^{-i}}^i(\tau \cdot s' | \tau \cdot s, b) = P_{\pi^{-i}}^i(s' | s, b), \quad (43)$$

$$r_{\pi^{-i}}^i(\tau \cdot s, b) = r_{\pi^{-i}}^i(s, b). \quad (44)$$

Thus the best-response MDP is H_i -state-symmetric with unchanged action labels.

Its Bellman optimality operator maps H_i -invariant value functions to H_i -invariant value functions by equation 43–equation 44. Starting value iteration from zero and using contraction therefore shows that the optimal value and optimal action-value functions are H_i invariant. Choosing a greedy action with one fixed tie-breaking order on $\mathcal{B}$ gives an H_i -invariant optimal stationary policy.

The H_i -orbits in $\mathcal{S}$ are exactly the fibers of z_i . Indeed, elements of H_i fix s^i and reorder the other coordinates; conversely, equal descriptors allow the remaining equal-valued coordinates to be matched by a permutation fixing i . Hence an H_i -invariant local policy is exactly a policy of the form $\chi \circ z_i$. The invariant optimal policy constructed above therefore proves equation 18.

It remains to compare agents. Fix i, j and choose $\sigma \in S_N$ with $\sigma(i) = j$. For any local policy ν^i , define its relabeling for agent j by

$$(\sigma_* \nu^i)(b | s') := \nu^i(b | \sigma^{-1} \cdot s'). \quad (45)$$

This is a bijection between the local policy spaces of agents i and j . Let

$$q := (\nu^i, \pi^{-i}), \quad q' := (\sigma_* \nu^i, \pi^{-j}).$$

For the deviating coordinate,

$$q'^{\sigma(i)}(b | \sigma \cdot s) = q^i(b | s)$$

by equation 45; for every $k \neq i$, the same identity holds between $q'^{\sigma(k)}$ and q^k by permutation invariance of π . Thus the trajectory relabeling used in Theorem 23, now between policies q and q' , gives

$$V_{q'}^j(\sigma \cdot s) = V_q^i(s).$$

Averaging over the permutation-invariant initial distribution and reindexing states yields

$$V_{\sigma_* \nu^i, \pi^{-j}}^j(\mu) = V_{\nu^i, \pi^{-i}}^i(\mu).$$

Taking maxima and using the bijectivity of $\nu^i \mapsto \sigma_* \nu^i$ shows that the optimal response values are equal for agents i and j . Similarly, equation 39 and equation 13 imply

$$V_{\pi}^j(\mu) = V_{\pi}^i(\mu).$$

The best-response gains are therefore equal. Combining this fact with equation 18 and the definition of the Nash gap proves equation 19. $\square$

I BLACK-BOX RATE TRANSFER FOR SHARED RULES

Proof of Theorem 12, part 3. Every solver iterate is pointed-orbit shared by hypothesis. Equation (19) therefore gives, for every fixed agent i and every iteration t ,

$$\text{Nash gap}_M(\pi_t) = \max_{\chi: \mathcal{Z} \rightarrow \Delta(\mathcal{B})} \left[V_{\chi \circ z_i, \pi_t^{-i}}^i(\mu) - V_{\pi_t}^i(\mu) \right].$$

Taking the minimum over $0 \leq t < T$ and applying equation 17 proves equation 20.

For the optional bound comparison, the unrestricted table has $D_{\text{full}} = N |\mathcal{C}|^N$ local policy blocks, while the shared table has $D_{\text{perm}} = |\mathcal{Z}|$. Orbit compatibility is precisely the additional hypothesis that the same certificate form controls the ordinary Nash gap before tying and the pointed-orbit gap after tying, using common auxiliary bounds. Both representations have state count $S = |\mathcal{C}|^N$. Therefore

$$\frac{B_{\text{perm}}(\epsilon)}{B_{\text{full}}(\epsilon)} = \frac{C_{\text{perm}}(\vartheta) S^\alpha D_{\text{perm}}^\delta \epsilon^{-p}}{C_{\text{full}}(\vartheta) S^\alpha D_{\text{full}}^\delta \epsilon^{-p}} \leq \left(\frac{|\mathcal{Z}|}{N |\mathcal{C}|^N} \right)^\delta.$$

This proves equation 21. The condition on ϑ is essential: for Euclidean shared-coordinate gradient methods, for example, smoothness may depend on the largest number of pointed states tied to one descriptor. The block count alone does not control that multiplicity. $\square$

J PROOF OF THE COMBINED REDUCTION

Proof of Theorem 16. The descended permutation action. Suppose $s, \tilde{s} \in \bar{s}$, so $\tilde{s} = L_g[s]$ for some $g \in G$. By equation 22,

$$\sigma \cdot \tilde{s} = \sigma \cdot L_g[s] = L_g[\sigma \cdot s].$$

Thus $\sigma \cdot \tilde{s}$ and $\sigma \cdot s$ belong to the same G orbit, so $\sigma \cdot \bar{s} := \rho(\sigma \cdot s)$ is independent of the representative. It is a group action because the original S_N action is one.

For quotient states $\bar{s}, \bar{s}'$ with representatives s, s' and any joint action a , reindex the target orbit by $u = \sigma \cdot v$ and use equation 11:

$$\begin{aligned} \bar{P}(\sigma \cdot \bar{s}' \mid \sigma \cdot \bar{s}, \sigma \cdot a) &= \sum_{u \in \sigma \cdot \bar{s}'} P(u \mid \sigma \cdot s, \sigma \cdot a) \\ &= \sum_{v \in \bar{s}'} P(\sigma \cdot v \mid \sigma \cdot s, \sigma \cdot a) \\ &= \sum_{v \in \bar{s}'} P(v \mid s, a) = \bar{P}(\bar{s}' \mid \bar{s}, a). \end{aligned}$$

The reward and potential identities follow directly:

$$\begin{aligned} \bar{r}^{\sigma(i)}(\sigma \cdot \bar{s}, \sigma \cdot a) &= r^{\sigma(i)}(\sigma \cdot s, \sigma \cdot a) = r^i(s, a) = \bar{r}^i(\bar{s}, a), \\ \bar{\phi}(\sigma \cdot \bar{s}, \sigma \cdot a) &= \phi(\sigma \cdot s, \sigma \cdot a) = \phi(s, a) = \bar{\phi}(\bar{s}, a). \end{aligned}$$

Finally, reindexing the orbit sum and using equation 13 gives

$$\bar{\mu}(\sigma \cdot \bar{s}) = \sum_{u \in \sigma \cdot \bar{s}} \mu(u) = \sum_{v \in \bar{s}} \mu(\sigma \cdot v) = \bar{\mu}(\bar{s}).$$

Hence $\bar{M}$ is permutation symmetric.

The two orbit orders agree. Commutativity makes the displayed formula in equation 23 a left action of $G \times S_N$. Define

$$\begin{aligned} F: \mathcal{W} &\longrightarrow (\mathcal{N} \times (\mathcal{S}/G))/S_N, \\ F([(i, s)]_{G \times S_N}) &:= [(i, \rho(s))]_{S_N}. \end{aligned}$$

The representative calculation above shows that F is well defined. It is surjective because every pointed quotient state has a representative in $\mathcal{N} \times \mathcal{S}$. If $F([(i, s)]) = F([(j, s')])$, there is a $\sigma \in S_N$ with $j = \sigma(i)$ and $\rho(s') = \rho(\sigma \cdot s)$. Therefore $s' = L_g[\sigma \cdot s]$ for some $g \in G$, so the two pointed states are in the same $G \times S_N$ orbit. Thus F is injective.

Similarly, commutativity induces a well-defined G action on $\mathcal{Z} = (\mathcal{N} \times \mathcal{S})/S_N$ by

$$g \cdot [(i, s)]_{S_N} := [(i, L_g[s])]_{S_N}.$$

Quotienting this orbit space by G identifies exactly the same pairs of pointed states as the product action. This proves both isomorphisms in equation 24.

Jointly invariant policies. If π is jointly invariant and $(j, s') = (\sigma(i), L_g[\sigma \cdot s])$, then

$$\begin{aligned} \pi^j(b \mid s') &= \pi^{\sigma(i)}(b \mid L_g[\sigma \cdot s]) \\ &= \pi^{\sigma(i)}(b \mid \sigma \cdot s) = \pi^i(b \mid s). \end{aligned}$$

Thus the local policy is constant on every combined pointed orbit and defines one rule on $\mathcal{W}$. Conversely, a rule on $\mathcal{W}$ is unchanged under the elements (g, e) and (e, σ) , so its induced policy is both G invariant and permutation invariant. This proves the policy correspondence.

Unrestricted best responses. Fix an agent i and let $H_i := \{\tau \in S_N : \tau(i) = i\}$. Against jointly invariant opponents, agent i 's best-response MDP is symmetric under $G \times H_i$. Indeed, reindexing the opponents' action sum by τ and applying first the permutation identities and then the state-only identities gives, for every $g \in G$, $\tau \in H_i$, and focal action b ,

$$\begin{aligned} P_{\pi-i}^i(L_g[\tau \cdot s'] \mid L_g[\tau \cdot s], b) &= P_{\pi-i}^i(s' \mid s, b), \\ r_{\pi-i}^i(L_g[\tau \cdot s], b) &= r_{\pi-i}^i(s, b). \end{aligned}$$

The focal action remains b because G fixes all action labels and $\tau(i) = i$. Value iteration from zero therefore yields a $G \times H_i$ -invariant optimal value and action-value function. Greedy selection with one fixed tie-breaking order gives a $G \times H_i$ -invariant optimal stationary response.

The $G \times H_i$ orbits of states are exactly the fibers of w_i . One implication follows immediately from equation 23. For the converse, if $w_i(s) = w_i(s')$, then some (g, σ) maps (i, s) to (i, s') ; hence $\sigma(i) = i$, so $\sigma \in H_i$ and $s' = L_g[\sigma \cdot s]$. The invariant optimal response is therefore of the form $\chi \circ w_i$. Finally, the relabeling argument in Theorem 24 still applies because a jointly invariant policy is permutation invariant. It shows that all agents have the same best-response gain. This proves equation 25.

Combined rate transfer. By the first part, the shared solver may be run on $\overline{M}$. Its pointed-state orbit space is identified with $\mathcal{W}$ by equation 24. Each iterate $\bar{\pi}_t$ is permutation invariant. For a fixed agent i , its best-response MDP is H_i symmetric, where $H_i = \{\tau \in S_N : \tau(i) = i\}$, by the same calculation as in Equations (43) and (44). The H_i orbits of $\overline{\mathcal{S}}$ are exactly the fibers of $z_i^{\overline{M}}$, and relabeling agents shows that all best-response gains are equal. Therefore

$$\text{OrbGap}_{\overline{M}}^i(\bar{\pi}_t) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

Lifting and equation 8 then give, at every iteration,

$$\text{Nash gap}_M(\bar{\pi}_t^\uparrow) = \text{Nash gap}_{\overline{M}}(\bar{\pi}_t).$$

Applying the assumed shared-solver certificate with $S = |\overline{\mathcal{S}}|$ and $D = |\mathcal{W}|$ proves equation 26.

Under the orbit-compatible monomial certificate, the full table has state count $|\mathcal{S}|$ and $N |\mathcal{S}|$ policy blocks, the permutation-only representation has state count $|\mathcal{S}|$ and $|\mathcal{Z}|$ blocks, the quotient-only representation has state count $|\overline{\mathcal{S}}|$ and $N |\overline{\mathcal{S}}|$ blocks, and the combined representation has state

count $|\overline{\mathcal{S}}|$ and $|\mathcal{W}|$ blocks. The assumed prefactor inequalities therefore give

$$\begin{aligned}\frac{B_{\text{both}}(\epsilon)}{B_{\text{full}}(\epsilon)} &\leq \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{N|\mathcal{S}|}\right)^\delta \\ &= \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|}\right)^\delta \left(\frac{|\mathcal{Z}|}{N|\mathcal{S}|}\right)^\delta, \\ \frac{B_{\text{both}}(\epsilon)}{B_{\text{perm}}(\epsilon)} &\leq \left(\frac{|\overline{\mathcal{S}}|}{|\mathcal{S}|}\right)^\alpha \left(\frac{|\mathcal{W}|}{|\mathcal{Z}|}\right)^\delta, \\ \frac{B_{\text{both}}(\epsilon)}{B_{\text{quot}}(\epsilon)} &\leq \left(\frac{|\mathcal{W}|}{N|\overline{\mathcal{S}}|}\right)^\delta.\end{aligned}$$

This proves Equations (27) to (29). The orbit isomorphisms identify $\mathcal{W}$ with $\mathcal{Z}/G$, so $|\mathcal{W}| \leq |\mathcal{Z}|$. Also, $\mathcal{W}$ is an orbit space of $\mathcal{N} \times \overline{\mathcal{S}}$, so $|\mathcal{W}| \leq N|\overline{\mathcal{S}}|$. If both stated G actions are free, the orbit-stabilizer theorem gives

$$|\overline{\mathcal{S}}| = \frac{|\mathcal{S}|}{|G|}, \quad |\mathcal{W}| = \frac{|\mathcal{Z}|}{|G|}.$$

These are separate assumptions: freeness on $\mathcal{S}$ need not imply freeness of the induced action on $\mathcal{Z}$, because $L_g[s]$ may agree with s after an agent relabeling. Therefore both are needed for the displayed multiplicative factors. The two equalities above show that $|\overline{\mathcal{S}}|/|\mathcal{S}|$ and $|\mathcal{W}|/|\mathcal{Z}|$ both equal $|G|^{-1}$. Substitution proves Equations (30) and (31). $\square$